1. **Write a program to check whether a number is even or odd using switch-case.**
   Code:
```
package Day2;
import java.util.Scanner;

public class EvenOddSwitch {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number: ");
        int num = sc.nextInt();

        switch(num % 2) {
            case 0:
                System.out.println("Even");
                break;
            default:
                System.out.println("Odd");
        }

        sc.close();
    }
}
```

2. **Write a program that takes a character as input and checks whether it is a vowel or consonant using**
   Code:
```
package Day2;
import java.util.Scanner;

public class VowelConsonantSwitch {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter character: ");
        char ch = Character.toLowerCase(sc.next().charAt(0));

        switch(ch) {
            case 'a':
            case 'e':
            case 'i':
            case 'o':
            case 'u':
                System.out.println("Vowel");
```

```
            break;
        default:
            System.out.println("Consonant");
    }

    sc.close();
  }
}
```

3. **Write a program that takes a grade (A, B, C, D, F) and displays the performance message**

Code:

```
package Day2;
import java.util.Scanner;

public class GradeMessage {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter grade: ");
    char grade = Character.toUpperCase(sc.next().charAt(0));

    switch(grade) {
      case 'A': System.out.println("Excellent"); break;
      case 'B': System.out.println("Very Good"); break;
      case 'C': System.out.println("Good"); break;
      case 'D': System.out.println("Pass"); break;
      case 'F': System.out.println("Fail"); break;
      default: System.out.println("Invalid Grade");
    }

    sc.close();
  }
}
```

4. **Create a menu-driven program for ATM operations:**
   - **1 → Check Balance**
   - **2 → Deposit**
   - **3 → Withdraw**
   - **4 → Exit**

Code:

```
package Day2;
```

```
import java.util.Scanner;

public class ATMMenu {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        double balance = 5000;

        System.out.println("1.Check Balance");
        System.out.println("2.Deposit");
        System.out.println("3.Withdraw");
        System.out.println("4.Exit");

        System.out.print("Enter choice: ");
        int choice = sc.nextInt();

        switch(choice) {
            case 1:
                System.out.println("Balance: " + balance);
                break;

            case 2:
                System.out.print("Enter deposit amount: ");
                double dep = sc.nextDouble();
                balance += dep;
                System.out.println("Updated Balance: " + balance);
                break;

            case 3:
                System.out.print("Enter withdraw amount: ");
                double wd = sc.nextDouble();
```

```
            if(wd <= balance) {
                balance -= wd;
                System.out.println("Updated Balance: " + balance);
            } else {
                System.out.println("Insufficient Balance");
            }
            break;

        case 4:
            System.out.println("Thank You");
            break;

        default:
            System.out.println("Invalid Choice");
    }

    sc.close();
  }
}
```

5. **Create a program where user selects food item number and the program displays item name and price.**

Code:

```
package Day2;
import java.util.Scanner;

public class FoodMenu {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.println("1. Burger");
    System.out.println("2. Pizza");
    System.out.println("3. Sandwich");

    System.out.print("Enter item number: ");
```

```
        int item = sc.nextInt();

        switch(item) {
            case 1: System.out.println("Burger - Rs150"); break;
            case 2: System.out.println("Pizza - Rs250"); break;
            case 3: System.out.println("Sandwich - Rs100"); break;
            default: System.out.println("Invalid Item");
        }

        sc.close();
    }
}
```

6. **Write a program to check whether a number is even or odd using switch-case.**

Code:

```
package Day2;

import java.util.Scanner;

public class EvenOddSwitch {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        switch (num % 2) {
            case 0:
                System.out.println("Even");
                break;

            case 1:
            case -1:
                System.out.println("Odd");
                break;

            default:
                System.out.println("Invalid");
        }

        sc.close();
    }
}
```

7. **Write a program that converts currency based on user choice (e.g., INR to USD, INR to EUR, etc.).**

Code:

```
package Day2;
import java.util.Scanner;

public class CurrencyConverter {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter INR amount: ");
        double inr = sc.nextDouble();

        System.out.println("1. USD");
        System.out.println("2. EUR");

        System.out.print("Choose currency: ");
        int choice = sc.nextInt();

        switch(choice) {
            case 1:
                System.out.println("USD: " + (inr / 83));
                break;
            case 2:
                System.out.println("EUR: " + (inr / 90));
                break;
            default:
                System.out.println("Invalid Choice");
        }

        sc.close();
    }
}
```

8. **Create a program that checks user role (Admin, User, Guest) and displays access permissions.**

Code:

```
package Day2;
import java.util.Scanner;

public class UserRole {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter role: ");
        String role = sc.nextLine();

        switch(role.toLowerCase()) {
            case "admin":
                System.out.println("Full Access");
                break;
            case "user":
                System.out.println("Limited Access");
                break;
            case "guest":
                System.out.println("View Only");
                break;
            default:
                System.out.println("Invalid Role");
        }

        sc.close();
    }
}
```

9. **Write a program to calculate the factorial of a given number using a while loop.**

Code:

```
package Day2;
import java.util.Scanner;

public class Factorial {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n, fact = 1;
        System.out.print("Enter number: ");
        n = sc.nextInt();

        while(n > 0) {
            fact *= n;
            n--;
        }

        System.out.println("Factorial = " + fact);
        sc.close();
    }
}
```

10. Write a program to print the multiplication table of a given number using a while loop.

Code:

```
package Day2;
import java.util.Scanner;

public class Table {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n, i = 1;
        System.out.print("Enter number: ");
        n = sc.nextInt();

        while(i <= 10) {
            System.out.println(n + " x " + i + " = " + (n*i));
            i++;
        }

        sc.close();
    }
}
```

11. Write a program to count the number of digits in a given number using a while loop.

Code:

```
package Day2;
import java.util.Scanner;

public class CountDigits {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n, count = 0;
        System.out.print("Enter number: ");
        n = sc.nextInt();

        while(n > 0) {
            count++;
            n /= 10;
        }

        System.out.println("Digits = " + count);
        sc.close();
    }
```

```
}
```

12. Write a program to reverse a given number using a while loop.

Code:

```
package Day2;
import java.util.Scanner;

public class ReverseNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n, rev = 0;
        System.out.print("Enter number: ");
        n = sc.nextInt();

        while(n > 0) {
            rev = rev * 10 + n % 10;
            n /= 10;
        }

        System.out.println("Reverse = " + rev);
        sc.close();
    }
}
```

13. Write a program to check whether a number is a palindrome using a while loop.

Code:

```
package Day2;
import java.util.Scanner;

public class PalindromeNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n, temp, rev = 0;
        System.out.print("Enter number: ");
        n = sc.nextInt();

        temp = n;

        while(n > 0) {
            rev = rev * 10 + n % 10;
            n /= 10;
        }
```

```
        if(temp == rev)
            System.out.println("Palindrome");
        else
            System.out.println("Not Palindrome");

        sc.close();
    }
}
```

14. Write a program to find the sum of digits of a number using a while loop.

Code:

```
package Day2;
import java.util.Scanner;

public class SumDigits {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n, sum = 0;
        System.out.print("Enter number: ");
        n = sc.nextInt();

        while(n > 0) {
            sum += n % 10;
            n /= 10;
        }

        System.out.println("Sum = " + sum);
        sc.close();
    }
}
```

15. **Write a program to check whether a number is an Armstrong number using a while loop.**

Code:

```
package Day2;
import java.util.Scanner;

public class Armstrong {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n, temp, sum = 0, rem;
```

```
        System.out.print("Enter number: ");
        n = sc.nextInt();

        temp = n;

        while(n > 0) {
            rem = n % 10;
            sum += rem * rem * rem;
            n /= 10;
        }

        if(temp == sum)
            System.out.println("Armstrong");
        else
            System.out.println("Not Armstrong");

        sc.close();
    }
}
```

16. Write a program to check whether a number is prime using a while loop.

Code:

```
package Day2;
import java.util.Scanner;

public class PrimeCheck {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n, i = 2;
        boolean prime = true;

        System.out.print("Enter number: ");
        n = sc.nextInt();

        while(i < n) {
            if(n % i == 0) {
                prime = false;
                break;
            }
            i++;
        }

        if(n <= 1)
```

```
            prime = false;

        if(prime)
            System.out.println("Prime");
        else
            System.out.println("Not Prime");

        sc.close();
    }
}
```

17. Write a program to print Fibonacci series up to N terms using a while loop.

Code:

```
package Day2;
import java.util.Scanner;

public class Fibonacci {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n, a = 0, b = 1, i = 1;

        System.out.print("Enter terms: ");
        n = sc.nextInt();

        while(i <= n) {
            System.out.print(a + " ");
            int c = a + b;
            a = b;
            b = c;
            i++;
        }

        sc.close();
    }
}
```